

Experiment No: 3

Title: Write a Python Program for the Water Jug Problem

Aim

To develop a Python program to solve the Water Jug Problem by determining the sequence of operations required to measure the desired quantity of water using two jugs of different capacities.

Objective

To implement the Water Jug Problem in Python using the operations of filling, emptying, and transferring water between the two jugs until the required quantity of water is obtained.

Algorithm

- **Step 1:** Start the program.
- **Step 2:** Read the capacities of Jug 1, Jug 2, and the target quantity from the user.
- **Step 3:** Check whether the target quantity can be achieved.
- **Step 4:** If the target is not possible, display an appropriate message and stop.
- **Step 5:** Fill Jug 1 with water.
- **Step 6:** Pour water from Jug 1 to Jug 2 and empty Jug 2 whenever it becomes full.
- **Step 7:** Repeat the process until the target quantity is obtained.
- **Step 8:** Display the steps and print the success message.

Program

```
from math import gcd
```

```
def water_jug(jug1, jug2, target):  
    if target > max(jug1, jug2):  
        print("Target cannot be achieved.")  
        return  
  
    if target % gcd(jug1, jug2) != 0:  
        print("Target cannot be achieved.")  
        return
```

```
x = 0
```

```
y = 0
```

```
print("\nSteps:")
```

```
while x != target and y != target:
```

```
    if x == 0:
```

```
        x = jug1
```

```
        print(f'Fill Jug1    -> ({x}, {y})")
```

```
    elif y == jug2:
```

```
        y = 0
```

```
        print(f'Empty Jug2   -> ({x}, {y})")
```

```
    else:
```

```
        transfer = min(x, jug2 - y)
```

```
        x -= transfer
```

```
        y += transfer
```

```
        print(f'Pour Jug1 -> Jug2 -> ({x}, {y})")
```

```
print("\nTarget achieved!")
```

```
jug1 = int(input("Enter capacity of Jug1: "))
```

```
jug2 = int(input("Enter capacity of Jug2: "))
```

```
target = int(input("Enter target quantity: "))
```

```
water_jug(jug1, jug2, target)
```

Sample Input

Enter capacity of Jug1: 4

Enter capacity of Jug2: 3

Enter target quantity: 2

Sample Output

```
Enter capacity of Jug1: 4
Enter capacity of Jug2: 3
Enter target quantity: 2
```

Steps:

```
Fill Jug1      -> (4, 0)
Pour Jug1->Jug2 -> (1, 3)
Empty Jug2     -> (1, 0)
Pour Jug1->Jug2 -> (0, 1)
Fill Jug1      -> (4, 1)
Pour Jug1->Jug2 -> (2, 3)
```

```
Target achieved!
```

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Result

Thus, the Python program for the **Water Jug Problem** was executed successfully, and the required target quantity of water was obtained.